



Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework 2: Rankine Bottoming Cycle

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I hereby certify that the lab report herein was completed in accordance with the McKelvey School of Engineering and Undergraduate Student academic integrity policies. I acknowledge that artificial intelligence (AI) tools were used to generate content included in this report.

Samantha Feldman

A handwritten signature in black ink that reads "Samantha Feldman".

Assumptions and Boundary Conditions

Assumptions

- The working fluid of the closed Rankine cycle is water.
- Possible damage of the turbine blades of the turbine blades via erosion from droplet impingement should be minimized, so the turbine exit quality has a lower limit.
- The condenser is cooled by water taken from the Colorado river water from the deep outlet of a major reservoir. The water will be pulled just downstream of a dam directly from a cold release point so as to remain at the coldest possible temperature from the source [1].
- Condenser supply piping can ignore minor losses and losses from the supply piping. Friction losses of the steam are also negligible in the condenser.
- The condenser coolant pipes are made of 90–10 copper–nickel, which provides strong corrosion resistance and is well suited for use with river water due to its excellent resistance to biological growth, erosion–corrosion, and chloride-induced degradation [2].
- The exit temperature of the Rankine cycle working fluid from the condenser will be slightly higher than the outlet temperature of the coolant. This is because heat transfer is not ideal, and the fluids do not reach complete thermal equilibrium before leaving the condenser.
- Assume a constant mass flow rate in the cycle.
- Assume an ideal pump as pump work is small compared to turbine work, the pump's inefficiency hardly shifts the cycle efficiency curve, and it simplifies the thermodynamic analysis.
- Assume the cooling water remains at atmospheric pressure because it is drawn directly from the river, and treat its density as constant over the temperature range of interest.
- Assume the pressure change of the coolant is very small, almost negligible, allowing C_p to be used to approximate the heat transfer (and corresponding power). The value of C_p is taken as that of the saturated fluid evaluated at the coolant outlet temperature.

- States $9 \rightarrow 6$ are assumed to be isentropic because the pump is modeled as an ideal pump. Similarly, the process from states $7 \rightarrow 8s$ is also isentropic.

Boundary Conditions

- Turbine isentropic efficiency (n_t) = 0.90.
- Turbine exit quality lower limit: $x = 0.90$ [3].
- The high-temperature heat exchanger (boiler) provides 30 MW of thermal energy: $\dot{Q}_H = 30$ MW.
- The pinch point for the heat exchanger that limits T_7 : $\Delta T_P = 12^\circ C$ [4].
- Gas turbine outlet temperature: $T_4 = 377.84^\circ C$.
- The condenser is a shell and tube heat exchanger with $N = 1000$ parallel tubes (number of parallel tubes), $D = 25\text{mm}$ internal diameter of each tube, and $L = 10\text{m}$ length (per pipe) [5].
- The condenser coolant inlet temperature from the Colorado river dam reservoir: $T_{cin} = 10^\circ C$.
- The maximum temperature rise of the coolant before discharge: $T_{cout} - T_{cin} = 3^\circ C$ [6].
- Condenser roughness for friction analysis: $\epsilon = 0.002\text{mm}$ [2].
- The pinch point between condenser Rankine outlet temperature and condenser coolant outlet temperature: $T_9 - T_{cout} = 3^\circ C$ [4].
- The coolant water density: $\rho = 999\text{kg}/\text{m}^3$.
- $C_P = 4.186\text{ kJ/kg}$.

Environmental Discussion

Environmentally, using coolant water from the Colorado River requires a careful balance. Under the Clean Water Act, thermal discharge standards limit the maximum temperature rise at discharge to about $3^\circ C$ above the ambient source water temperature [6]. Because deep reservoir intake water is typically around $10^\circ C$ due to thermal stratification, the allowable coolant exit temperature is therefore approximately $13^\circ C$.

Controlling discharge temperature is critical because warmer return water reduces dissolved oxygen levels, which can decrease wildlife survival and harm invertebrates. Since the Colorado River supports species adapted to relatively cold conditions, sustained increases toward 13°C can stress native species, promote invasive species, and contribute to ecosystem imbalance [7].

Rankine Cycle Diagrams

Figure 1 displays the schematic of a Combined Cycle Gas Turbine (CCGT). In this design homework, only the bottoming cycle is being analyzed.

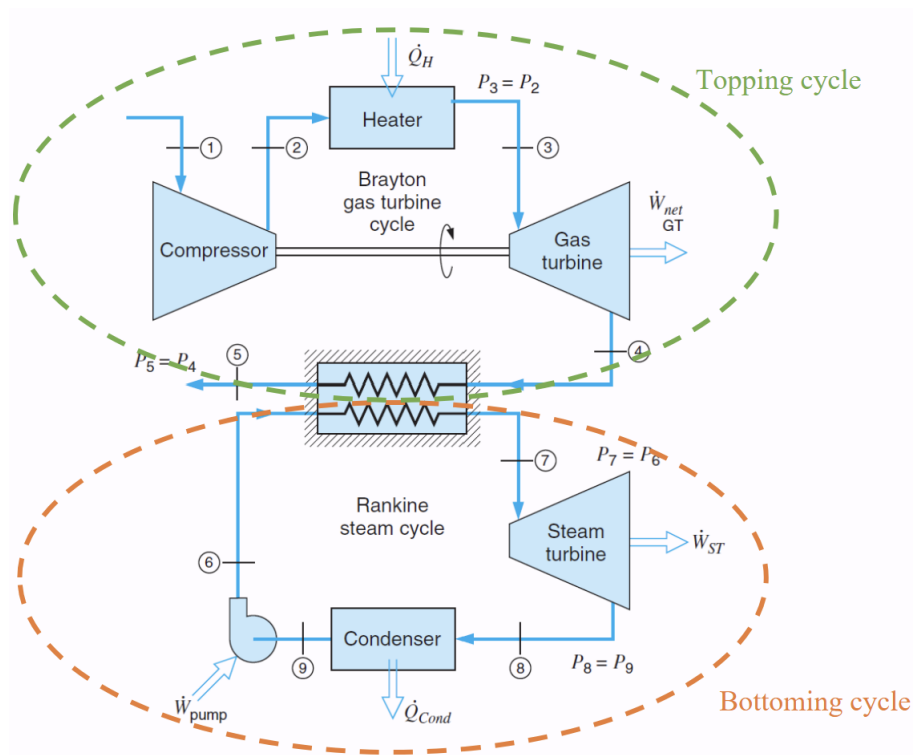


Figure 1 Schematic of a Combined Cycle Gas Turbine (CCGT).

Figure 2 shows a graphical representation of the relationship between temperature and entropy for the 4-stage Rankine cycle. The state numbers correspond to the state numbers displayed in Fig 1.

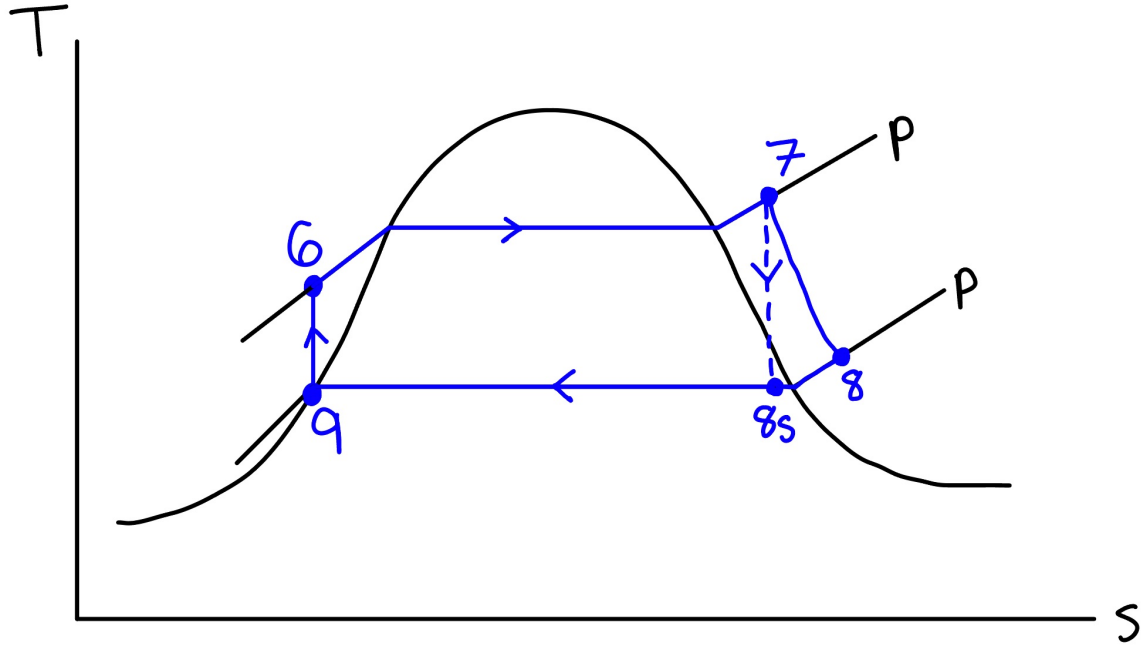


Figure 2 T-S Diagram of the Rankine Cycle.

Equations

The guiding equation for this design is the thermal efficiency (η_{th}) of the Rankine cycle which is defined in the equation below.

$$\eta_{th} = \frac{w_t - w_p - w_{p,cl}}{q_H} \quad (1)$$

Where w_t is the specific work of the turbine [kJ/kg], w_p is the specific work of the Rankine pump [kJ/kg], $w_{p,cl}$ is the specific work of the condenser coolant pump [kJ/kg], and q_H is the specific thermal energy provided by the high temperature heat exchanger (boiler).

By isolating the turbine and applying an energy balance, the specific turbine work can be determined. The isentropic turbine efficiency relation is then used to obtain the actual turbine exit enthalpy. With the exit enthalpy known, all quantities required to evaluate the turbine exit quality become available. Finally, the following equations provide the temperature at the turbine inlet.

$$w_t = h_7 - h_8 \quad (2)$$

$$h_8 = h_7 - \eta_t(h_7 - h_{8s}) \quad (3)$$

$$x_8 = \frac{h_8 - h_{8f}}{h_{8g} - h_{8f}} \quad (4)$$

$$T_7 = T_4 - \Delta T_p \quad (5)$$

Where w_t is the specific work of the turbine [kJ/kg], h_7 and h_8 are the enthalpy [kJ/kg] entering and exiting the turbine respectively, and h_{8s} is the ideal turbine exit enthalpy [kJ/kg]. x_8 is the turbine exit quality, h_{8f} and h_{8g} is the saturated fluid and gas enthalpy [kJ/kg] at the turbine exit pressure respectively. T_7 is the temperature [Celsius] entering the steam turbine and T_4 is the temperature [Celsius] exiting the gas turbine.

Isolating the heat exchanger, doing an energy balance, and isolating the mass flow rate solves for the Rankine cycle mass flow rate.

$$\dot{m} = \frac{\dot{Q}_H}{h_7 - h_6} \quad (6)$$

Where $\dot{m}$ is the Rankine cycle mass flow rate [kg/s], h_7 is the enthalpy [kJ/kg] exiting the heat exchanger, and h_6 is the enthalpy [kJ/kg] entering the heat exchanger $\dot{Q}_{cond}$ in kW. Now that the flow rate is known, the specific input heat energy can be found with the equation below.

$$q_H = \frac{\dot{Q}_H}{\dot{m}} \quad (7)$$

Isolating the condenser and performing an energy balance solves for the heat energy exiting the condenser.

$$\dot{Q}_{cond} = \dot{m}(h_8 - h_9) \quad (8)$$

Where $\dot{m}$ is the Rankine cycle mass flow rate [kg/s], h_8 is the enthalpy entering the condenser, and h_9 is the condenser exit enthalpy [kJ/kg].

Assuming the change in pressure in the coolant tubes is incredibly small to the point where it is negligible another energy balance can be used to isolate the condenser fluid mass flow rate by plugging in eq. 8.

$$h_8 - h_9 = c_p(T_8 - T_9)$$

$$\dot{m}_c = \frac{\dot{Q}_{cond}}{c_p(T_8 - T_9)} \quad (9)$$

Where $\dot{m}_c$ is the coolant mass flow rate [kg/s], c_p is specific heat capacity at a constant pressure [kJ/kg-K], T_8 is condenser entrance temperature [K], and T_9 is the condenser exit temperature [K].

Isolating the coolant flow can attain the coolant pump power using the Bernoulli equation and work definition.

$$p_8 + \frac{1}{2}\alpha_8\rho v_8^2 + \rho g z_8 = p_9 + \frac{1}{2}\alpha_9\rho v_9^2 + \rho g z_9 + \sum f \frac{l\rho v^2}{2D} \quad (10)$$

Where p is the pressure [Pa], α is the kinetic energy correction factor, ρ is the density [kg/m³], g is gravity (9.81 m/s²), z is height [m], f is the pipe roughness [mm], l is the length of a pipe [m], D is the internal pipe diameter [mm], v is velocity [m/s²]. The velocity difference between the entrance and exit of the coolant tubes is the same, as is the height. Eq. 10 is simplified below.

$$p_8 - p_9 = \Delta p = f \frac{l\rho v^2}{2D} \quad (11)$$

To solve for the average velocity in the pipes the following calculations are done.

$$A = \frac{\pi D^2}{4} \quad (12)$$

$$\dot{m}_c = \rho AV$$

$$v = \frac{\dot{m}_c}{N\rho A} \quad (13)$$

Where v is the average velocity [m/s^2] in the coolant pipes, and N is the number of pipes, and A is the cross sectional area [m^2] of one pipe. To find the specific work of the coolant pump the following equation is used.

$$w_{p,cl} = \int v dp = v \Delta p \quad (14)$$

$$v = 1/\rho \quad (15)$$

where v is the specific volume of the water [m^3/kg], $w_{p,cl}$ is the specific work [kJ/kg] of the coolant pump, and ρ is the density of water [kg/m^3].

The last piece to solve for the thermal efficiency is finding the specific work of the Rankine cycle pump through isolation and an energy balance.

$$w_p = h_6 - h_9 \quad (16)$$

Where w_p is the specific work [kJ/kg] of the Rankine cycle pump, h_6 is the pump exit enthalpy [kJ/kg], and h_9 is the pump entrance enthalpy [kJ/kg].

To calculate the overall power, the turbine output power, Rankine pump input power, coolant pump input power are calculated then summed using eq 2, 6, 9, 14, and 16.

$$\dot{W}_t = w_t * \dot{m} \quad (17)$$

$$\dot{W}_p = w_t * \dot{m} \quad (18)$$

$$\dot{W}_{p,cl} = w_{p,cl} * \dot{m}_c \quad (19)$$

$$\dot{W}_{net} = \dot{W}_t - \dot{W}_p - \dot{W}_{p,cl} \quad (20)$$

Where $\dot{W}_t$ is the turbine output power [W], $\dot{W}_p$ is the Rankine cycle pump input power [W], $\dot{W}_{p,cl}$ is the coolant pump input power, and $\dot{W}_{net}$ is the net power of the bottoming cycle.

Additionally, to determine the overall CCGT thermodynamic efficiency for both the topping and bottoming cycles, it is important to note that the bottoming (Rankine) cycle utilizes the waste heat rejected by the topping (Brayton) cycle. Therefore, the overall efficiency $\eta_{overall}$ is given by the equation below.

$$\eta_{overall} = 1 - (1 - \eta_B)(1 - \eta_R) \quad (21)$$

Where η_B is the topping Brayton cycle thermal efficiency and η_R is the Rankin bottoming cycle thermal efficiency.

RESULTS

Figure 3 shows thermal efficiency (left y-axis) and turbine exit quality (right y-axis) as functions of turbine inlet pressure, illustrating how changes in inlet pressure influence both performance optimization and outlet steam condition.

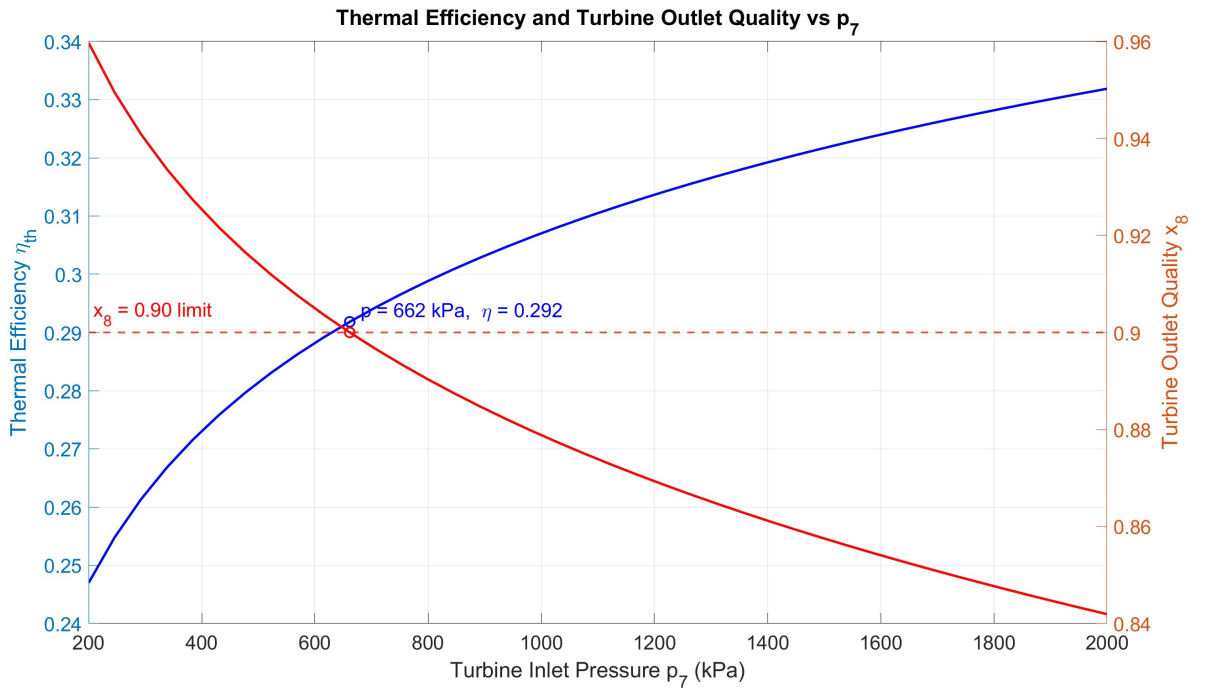


Figure 3 Thermal efficiency (left y-axis) and turbine exit quality (right y-axis) as functions of turbine inlet pressure.

The minimum turbine exit quality after performing a literature search is $x_8 = 0.90$ [3]. If the quality were to fall below 90% the Rankine steam cycle performance would start to deteriorate due to possible damage of the turbine blades via erosion from droplet impingement. As the thermal efficiency follows a positive trend in the plot and the turbine exit quality follows a negative trend, the thermal efficiency should be at the highest possible inlet pressure, however that is capped by the minimum x_8 . Based on this the optimal thermal efficiency as at $\eta = 0.292$ at a turbine inlet pressure (p_7) of 662 kPa at $x_8 = 0.90$.

Table 1 lists the values at the optimized layout of the Rankine bottoming cycle.

Table 1 Optimized Rankine Cycle Values

Value	Value
Rankine Cycle thermodynamic efficiency	29.2%
Turbine inlet pressure (p_7)	662 kPa
Rankine cycle mass flow rate	9.58 kg/s
Coolant loop mass flow rate	1691.7645 kg/s
Turbine power output	8761.15 kW
Rankine cycle pump power input	6.33 kW
Coolant loop pump power input	8.05 kW
Total net power output	8746.77 kW
Total thermodynamic efficiency of the entire CCGT (Topping and bottoming cycle)	51.48%

Discussion and Sanity Check

Generally, a Rankine cycle thermodynamic efficiency is around the 30% range, so the calculated value of 29.2% is consistent with typical performance. Although the coolant loop mass flow rate of 1691.76 kg/s may initially seem large, it is realistic for an industrial-scale Rankine cycle condenser [8]. This is particularly true for once-through cooling systems drawing from a large water source such as the Colorado River, where discharge temperatures are limited to a maximum increase of 3°C under Clean Water Act constraints [6]. Given these regulatory limits and the scale of operation, such

a coolant flow rate is reasonable for an industrial condenser.

The net power output of 8.75 MW is also within a realistic range for an industrial Rankine cycle, especially when compared to the boiler heat input of 30 MW. Similarly, the Rankine cycle working fluid mass flow rate of 9.58 kg/s is on the order of magnitude expected for a small industrial system, and the turbine inlet pressure of 662 kPa falls within the low-to-moderate range typical for such applications [9].

Finally, the overall thermodynamic efficiency of the combined cycle gas turbine (CCGT) system, calculated as 51.48%, is of a typical magnitude for modern combined-cycle power plants [10].

References

- [1] 2026, “Colorado River,” Wikipedia, accessed March 1, 2026, https://en.wikipedia.org/wiki/Colorado_River?utm_source=chatgpt.com
- [2] 2003, “Surface Roughness Coefficients,” The Engineering ToolBox, accessed March 1, 2026, https://www.engineeringtoolbox.com/surface-roughness-ventilation-ducts-d_209.html
- [3] Murugan, R. S. and Subbarao, P. V., 2008, “Thermodynamic Analysis of Rankine-Kalina Combined Cycle,” Int. J. of Thermodynamics, accessed March 1, 2026, <https://dergipark.org.tr/tr/download/article-file/65735>
- [4] Ahmed, A., Esmaeil, K. K., Irfan, M. A., and Al-Mufadi, F. A., 2018, “Design Methodology of Heat Recovery Steam Generator in Electric Utility for Waste Heat Recovery,” International Journal of Low-Carbon Technologies, accessed March 1, 2026, <https://academic.oup.com/ijlct/article/13/4/369/5095657?login=true>
- [5] 2026, “MEMS 3430 - Design of Thermal Systems DH 2: Rankine Bottoming Cycle,” Washington University in St. Louis.
- [6] 1988, “Water Quality Standards Criteria Summaries,” Environmental Protection Agency, accessed March 1, 2026, [https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockkey=00001NPM.TXT#:~:text=During%20reproductive%20seasons%20\(generally%20April,on%20a%20site%20specific%20basis.](https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockkey=00001NPM.TXT#:~:text=During%20reproductive%20seasons%20(generally%20April,on%20a%20site%20specific%20basis.)
- [7] 2024, “Question of the Month: What are the impacts of warmer lake temperatures,” 30 Mile River Watershet Association, accessed March 1, 2026, <https://30mileriver.org/question-of-the-month-what-are-the-impacts-of-warmer-lake-temperatures/>
- [8] Ravindran, R. V., Cotter, D., Wilson, C., Huang, M. J., and Hewitt, N. J., 2024, “Experimental investigation of a small-scale reversible high temperature heat pump - organic Rankine cycle system for industrial waste heat recovery,” Applied Thermal Engineering, accessed March 1, 2026, <https://www.sciencedirect.com/science/article/pii/S1359431124019057>
- [9] 2026, “Taylor Made and Flexible Industrial Steam Turbines,” Siemens Energy Global, accessed March 3, 2026, <https://www.siemens-energy.com/global/en/home/products-services/product/industrial-steam-turbines.html>
- [10] 2021, “Gas Turbine Combined Cycle,” Science Direct, accessed March 1, 2026, <https://www.sciencedirect.com/topics/engineering/gas-turbine-combined-cycle#:~:text=NGCC%2C%20or%20gas%20turbine%20combined,or%20more%20through%20this%20combination.&text=How%20useful%20is%20this%20definition?>

A Acknowledgment of Artificial Intelligence Resources

AI tools were used to look up LaTeX and Matlab code and was verified by inputting into Overleaf and Matlab respectively. Additionally, AI tools were used as a starting point for literature research, all information included from the output was verified by consulting the source. Anything not verified was not included in the report. A list of the AI prompts used is included below.

Cited AI Prompts

- ChatGPT, response to “between the mojave river, colorado river, lake mead, and lake havasu which would be the best coolant for a rankine cycle condensor for solar power plant in mojave dessert” OpenAI, February 28, 2026.
- ChatGPT, response to ”provide approximate intake temperature for colorado river cold water and lake mead from deep intake” OpenAI, February 28, 2026.
- Gemini, response to “pinch point range for steam turbine industrial rankine cycle for heat exchanger” Google, February 29, 2026.
- Copilot, response to “Take existing Matlab code and have it print these values: wt, wp at $x = 0.9$ ” Microsoft, March 3, 2026.
- Copilot, response to “Why is this latex code throwing an error” Microsoft, March 5, 2026.

B MATLAB Code

DHW2 Code

```
1 clear; clc;
2
3 %% === Font Controls ===
4 axisFont    = 16;    % controls axis labels, tick numbers, title
5 legendFont  = 14;    % controls legend text (if added later)
6
7 %% === Inputs ===
8 nt = 0.9;          % turbine isentropic efficiency
9 Qhdot = 30e3;       % input heat energy in kW
10 N = 1000;          % number of parallel tubes
11 d = 0.025;         % internal diameter (m)
12 l = 10;            % tube length (m)
13 ΔTp = 12;          % pinch point (C)
14 t4 = 377.84;        % turbine inlet temperature (C)
15 Tcin = 10;          % coolant inlet (C)
16 Tcout = Tcin + 3;    % coolant outlet (C)
17 t9 = Tcout + 3;      % condenser outlet (C)
18 f = 0.002;          % roughness
19 t7 = t4 - ΔTp; % turbine inlet temperature after pinch
20 x8min = 0.90;        % quality limit
21
22 %% === Pressure Sweep ===
23 P7_range = linspace(200, 2000, 40); % kPa
24 eta_vals = zeros(size(P7_range));
25 x8_vals  = zeros(size(P7_range));
26
27 wt_vals  = zeros(size(P7_range));
28 wp_vals  = zeros(size(P7_range));
29 wpcl_vals = zeros(size(P7_range));
30 mdot_vals = zeros(size(P7_range));
31 mcdot_vals = zeros(size(P7_range));
32 qh_vals  = zeros(size(P7_range));
33
```

```

34 %% === Loop Over Pressures ===
35 for j = 1:length(P7_range)
36
37     p7 = P7_range(j);          % kPa
38     p7b = p7/100;              % bar
39
40     % --- State 7 ---
41     h7 = XSteam('h_pt', p7b, t7);
42     s7 = XSteam('s_pt', p7b, t7);
43
44     % --- Expansion 7      8 ---
45     p8b = XSteam('psat-T', t9);
46     p8   = p8b*100;
47
48     h8s = XSteam('h-ps', p8b, s7);
49     h8   = h7 - nt*(h7 - h8s);
50
51     h8f = XSteam('hL-p', p8b);
52     h8g = XSteam('hV-p', p8b);
53
54     % --- Quality ---
55     x8_vals(j) = (h8 - h8f) / (h8g - h8f);
56
57     % --- Turbine work ---
58     wt = h7 - h8;
59
60     % --- Pump work ---
61     v9 = XSteam('vL-p', p8b);
62     wp = v9*(p7 - p8);
63
64     % --- State 6 ---
65     h9 = XSteam('hL-p', p8b);
66     h6 = h9 + v9*(p7 - p8);
67
68     % --- Mass flow ---
69     mdot = Qhdot/(h7 - h6);
70     qh    = Qhdot/mdot;
71

```

```

72     % --- Condenser heat ---
73     Qdotc = mdot*(h8 - h9);
74
75     % --- Coolant loop ---
76     cp = 4.186;
77     mcdot = Qdotc/(cp*(Tcout - Tcin));
78
79     rho = 999;
80     A = (pi*d^2)/4;
81     vel = mcdot/(N*rho*A);
82
83     Δp = (f*l*rho*(vel^2))/(2*d);
84     v_spec = 1/999;
85     wpcl = v_spec*(Δp/1000);
86
87     % --- Thermal efficiency ---
88     eta_vals(j) = (wt - wp - wpcl)/qh;
89
90     wt_vals(j) = wt;
91     wp_vals(j) = wp;
92     wpcl_vals(j) = wpcl;
93     mdot_vals(j) = mdot;
94     mcdot_vals(j) = mcdot;
95     qh_vals(j) = qh;
96
97 end
98
99 %% === Plot Efficiency + Quality ===
100 figure;
101
102 yyaxis left
103 plot(P7_range, eta_vals, 'b-', 'LineWidth', 2);
104 ylabel('Thermal Efficiency \eta_{th}', 'FontSize', axisFont);
105 xlabel('Turbine Inlet Pressure p_7 (kPa)', 'FontSize', axisFont);
106 title('Thermal Efficiency and Turbine Outlet Quality vs p_7', 'FontSize', ...
       axisFont);
107 set(gca, 'FontSize', axisFont);
108 grid on;

```

```

109
110 yyaxis right
111 plot(P7_range, x8_vals, 'r-', 'LineWidth', 2);
112 ylabel('Turbine Outlet Quality x_8', 'FontSize', axisFont);
113 set(gca, 'FontSize', axisFont);
114
115 %% === Mark Quality Limit x = 0.90 ===
116 x_limit = 0.90;
117
118 % Find closest index
119 [~, idx_limit] = min(abs(x8_vals - x_limit));
120
121 p_limit = P7_range(idx_limit);
122 eta_limit = eta_vals(idx_limit);
123
124 % Horizontal line at x = 0.90
125 yyaxis right
126 yl = yline(x_limit, '--r', 'x_8 = 0.90 limit', 'LineWidth', 1.5);
127 yl.LabelHorizontalAlignment = 'left';
128 yl.FontSize = axisFont;
129
130 % Mark point on quality curve
131 hold on;
132 plot(p_limit, x8_vals(idx_limit), 'ro', 'MarkerSize', 8, 'LineWidth', 1.5);
133
134 % Mark corresponding point on efficiency curve
135 yyaxis left
136 plot(p_limit, eta_limit, 'bo', 'MarkerSize', 8, 'LineWidth', 1.5);
137
138 % Annotate
139 text(p_limit, eta_limit, sprintf(' p = %.0f kPa, \eta = %.3f', p_limit, ...
    eta_limit), ...
    'VerticalAlignment', 'bottom', 'Color', 'b', 'FontSize', axisFont);
140
141
142 %% === Extract and Print Values at x8 = 0.90 ===
143 x_target = 0.90;
144
145 % Find index where quality is closest to 0.90

```

```

146 [~, idx] = min(abs(x8_vals - x_target));
147
148 % Extract corresponding pressure
149 p_target = P7_range(idx);
150
151 % Extract stored values
152 wt      = wt_vals(idx);
153 wp      = wp_vals(idx);
154 wpcl    = wpcl_vals(idx);
155 mdot    = mdot_vals(idx);
156 mcdot   = mcdot_vals(idx);
157 qh      = qh_vals(idx);
158 eta     = eta_vals(idx);
159 x8      = x8_vals(idx);
160
161 % Compute powers
162 Wturbine = wt * mdot;
163 Wpump    = wp * mdot;
164 Wcool    = wpcl * mcdot;
165 Wnet     = Wturbine - Wpump - Wcool;
166
167 % Compute Overall total system (Brayton + Rankine) Thermo efficiency
168 etab = .3148; % Topping Brayton cycle efficiency
169 etao = etab + eta - (etab*eta); %overall efficiency
170 etaocheck = 1 - (1- etab)*(1-eta); %overall efficiency check
171
172 %% Print results
173 fprintf('\n== Rankine Cycle Results at x8 = %.2f ==\n', x_target);
174 fprintf('Corresponding turbine inlet pressure p7: %.1f kPa\n', p_target);
175 fprintf('Turbine exit quality x8:                %.4f\n', x8);
176 fprintf('Mass flow rate (mdot):                    %.4f kg/s\n', mdot);
177 fprintf('Coolant mass flow rate (mcdot):            %.4f kg/s\n', mcdot);
178 fprintf('Turbine power output:                        %.2f kW\n', Wturbine);
179 fprintf('Rankine pump power input:                      %.2f kW\n', Wpump);
180 fprintf('Coolant pump power input:                      %.2f kW\n', Wcool);
181 fprintf('Total net power output:                       %.2f kW\n', Wnet);
182 fprintf('Thermodynamic efficiency (eta):                %.4f\n\n', eta);
183 fprintf('Overall Thermodynamic efficiency (etao):       %.4f\n\n', etao);

```

```
184 fprintf('Overall Thermodynamic efficiency check (etaocheck):      ...  
    %.4f\n\n', etao);
```